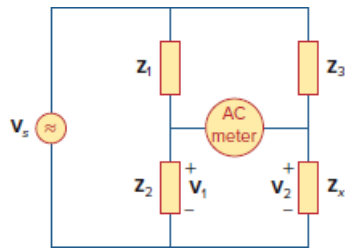


Problem

The ac bridge in Fig. 9.37 is balanced when $R_1 = 400\ \Omega$, $R_2 = 600\ \Omega$, $R_3 = 1.2\ \text{k}\Omega$, and $C_2 = 0.3\ \mu\text{F}$. Find R_x and C_x . Assume R_2 and C_2 are in series.

Figure 9.37



Step-by-step solution

Step 1 of 3

Consider the general ac bridge circuit.

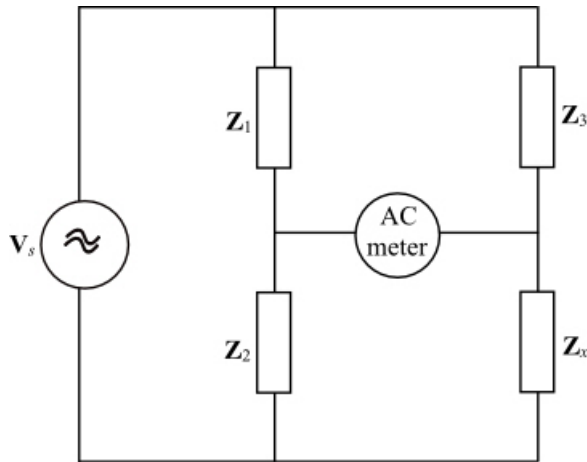


Figure 1: A general ac bridge

Step 2 of 3

The balanced equation for the ac bridge circuit is,

$$Z_x = \frac{Z_3}{Z_1} Z_2$$

Here,

$$Z_1 = R_1$$

$$Z_2 = R_2 + \frac{1}{j\omega C_2}$$

$$Z_3 = R_3$$

$$Z_x = R_x + \frac{1}{j\omega C_x}$$

Step 3 of 3

Substitute $400\ \Omega$ for R_1 , $600 + \frac{1}{j\omega(0.3 \times 10^{-6})}\ \Omega$ for Z_2 , and $1.2 \times 10^3\ \Omega$ for Z_3 in

balanced equation.

$$Z_x = \frac{(1.2 \times 10^3)}{400} \left(600 + \frac{1}{j\omega(0.3 \times 10^{-6})} \right)$$

$$R_x + \frac{1}{j\omega C_x} = 3 \left(600 + \frac{1}{j\omega(0.3 \times 10^{-6})} \right)$$

$$R_x + \frac{1}{j\omega C_x} = 1800 + \frac{1}{j\omega(0.1 \times 10^{-6})}$$

Equate real and imaginary parts. So, the real part is,

$$R_x = 1800\ \Omega$$

$$= 18\ \text{k}\Omega$$

The Imaginary part is,

$$\frac{1}{j\omega C_x} = \frac{1}{j\omega(0.1 \times 10^{-6})}$$

$$C_x = 0.1 \times 10^{-6}\ \text{F}$$

$$C_x = 0.1\ \mu\text{F}$$

Thus, the value of resistance R_x at which bridge balances is $18\ \text{k}\Omega$ and capacitance C_x is

$$0.1\ \mu\text{F}$$